

DUMBRAVENI, RomaniaWMO: **151890**Lat: **46.2278N**Lon: **24.5917E**Elev: **1047**StdP: **14.15**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.2	9.8	1.4	6.1	5.0	7.2	8.3	10.8	16.5	39.4	13.7	35.4	1.0	70	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	25.6	89.0	69.6	86.3	68.9	83.5	67.9	72.2	84.4	70.7	82.5	69.3	80.3	3.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.6	109.1	76.4	67.0	103.4	74.7	65.7	98.5	73.3	36.6	84.5	35.2	82.3	34.0	80.3	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.7	11.8	9.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	27.6	34.5	40.7	50.3	60.3	66.9	68.1	69.4	60.0	50.8	41.5	32.3
	DBStd	15.80	9.00	7.88	7.37	8.15	5.53	5.21	4.77	4.94	7.40	6.60	7.73	7.14
	HDD50	2428	694	435	296	97	4	0	0	0	8	74	270	550
	HDD65	5798	1159	855	755	442	161	38	26	18	181	442	706	1015
	CDD50	2526	0	0	7	106	324	507	562	600	308	97	15	0
	CDD65	419	0	0	0	0	17	95	123	153	31	0	0	0
	CDH74	4764	0	0	2	81	305	1011	1151	1701	488	25	0	0
	CDH80	1521	0	0	0	3	52	295	360	673	138	0	0	0
	WSAvg	3.2	2.7	3.2	3.7	4.2	3.6	3.4	3.4	2.9	3.1	2.9	2.6	2.9
	PrecAvg	25.3	1.0	0.9	1.4	2.1	3.1	4.0	3.3	2.4	2.4	2.0	1.4	1.3
(15) Precipitation	PrecMax	32.8	2.2	1.9	2.4	3.6	6.4	8.1	5.5	5.9	5.8	3.8	3.1	2.6
	PrecMin	17.9	0.2	0.1	0.2	0.5	1.2	0.7	1.1	0.5	0.2	0.2	0.0	0.2
	PrecStd	4.2	0.5	0.4	0.6	0.9	1.3	2.0	1.1	1.4	1.3	1.0	0.7	0.7
	PrecStd	4.2	0.5	0.4	0.6	0.9	1.3	2.0	1.1	1.4	1.3	1.0	0.7	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.1	59.3	69.5	80.0	85.4	89.4	91.2	94.2	88.3	76.7	67.9	51.2
		MCWB	43.7	50.6	53.6	59.4	62.6	71.6	71.7	70.6	66.4	60.6	56.7	46.5
	2%	DB	44.0	54.7	63.4	76.5	81.4	86.7	87.3	89.8	83.6	73.6	61.5	46.4
		MCWB	40.7	47.0	50.3	58.8	63.5	70.1	70.7	68.9	65.4	58.8	53.7	43.8
	5%	DB	40.3	49.7	57.9	71.5	77.7	83.5	84.1	86.8	79.7	68.9	57.3	43.3
		MCWB	38.0	44.3	47.6	56.2	62.1	68.9	69.1	68.8	63.3	58.2	51.5	41.5
	10%	DB	37.1	45.3	53.5	66.8	73.7	80.1	80.5	83.7	75.1	63.5	53.3	40.5
		MCWB	35.6	41.4	45.1	53.9	60.3	67.3	67.9	68.0	62.6	55.4	48.8	39.2
	0.4%	WB	44.5	51.5	54.5	61.7	69.2	74.4	74.3	73.6	69.9	64.5	57.1	47.2
		MCDB	47.9	59.9	68.4	76.3	80.1	85.7	85.0	87.6	81.6	72.0	65.0	51.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.2	47.6	51.5	59.6	66.4	71.8	71.9	71.8	67.1	60.9	54.5	43.9
		MCDB	43.9	53.6	61.6	73.3	75.5	82.9	83.5	85.1	78.7	70.3	60.8	45.9
	5%	WB	38.2	44.8	48.7	57.7	64.1	70.0	70.4	70.2	65.1	58.8	52.6	41.6
		MCDB	40.0	48.9	57.0	70.0	73.5	80.8	81.2	82.8	77.0	66.8	57.1	43.0
	10%	WB	36.0	41.8	46.0	55.3	61.9	68.3	69.0	68.8	63.2	56.2	49.7	39.3
		MCDB	37.0	44.5	52.3	64.4	70.7	78.0	78.8	81.0	73.0	62.6	53.5	40.3
(35) Mean Daily Temperature Range	5% DB	MDBR	14.0	17.4	20.6	23.3	21.9	22.9	22.8	25.6	23.6	22.9	17.5	10.8
		MCDBR	15.2	23.5	30.0	33.1	29.8	28.0	28.6	31.6	31.8	32.7	24.5	15.8
		MCWBR	11.5	16.3	17.8	16.4	13.6	12.1	12.7	12.5	13.7	18.9	16.3	12.5
	5% WB	MCDBR	14.5	23.5	28.4	30.6	26.2	25.3	25.5	28.9	28.8	29.1	22.7	14.7
		MCWBR	11.3	16.2	16.9	15.7	13.6	11.9	11.9	12.5	13.0	17.4	15.2	11.7
(40) Clear-Sky Solar Irradiance	taub		0.313	0.341	0.382	0.437	0.419	0.415	0.435	0.442	0.400	0.363	0.340	0.314
	taud		2.330	2.285	2.236	2.110	2.227	2.271	2.222	2.199	2.282	2.375	2.387	2.357
	Ebn at Noon		245	260	265	259	270	271	263	255	256	250	233	231
	Edh at Noon		24	31	38	47	43	42	43	43	36	28	23	22
	RadAvg		420	688	1036	1376	1695	1836	1831	1676	1203	827	470	325
(45) All-Sky Solar Radiation	RadStd		51	78	117	140	157	158	155	148	130	102	56	40

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days	
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